

### Getting Started

Start a new repo:  
git init

Clone an existing repo:  
git clone <url>

### Prepare to Commit

Add untracked file or unstaged changes:  
git add <file>

Add all untracked files and unstaged changes:  
git add .

Choose which parts of a file to stage:  
git add -p

Move file:  
git mv <old> <new>

Delete file:  
git rm <file>

Tell Git to forget about a file without deleting it:  
git rm --cached <file>

Unstage one file:  
git reset <file>

OR  
git restore --staged <file>

Unstage everything:  
git reset

OR  
git restore --staged .

Check what you added:  
git status

### Make Commits

Make a commit (and open text editor to write message):  
git commit

Make a commit:  
git commit -m 'message'

Commit all unstaged changes:  
git commit -am 'message'

### Move Between Branches

Switch branches:  
git switch <name>

OR  
git checkout <name>

Create a branch:  
git switch -c <name>

OR  
git checkout -b <name>

List branches:  
git branch

List branches by most recently committed to:  
git branch --sort=-committerdate

Delete a branch:  
git branch -d <name>

Force delete a branch:  
git branch -D <name>

### Diff Staged/Unstaged Changes

Diff all staged and unstaged changes:  
git diff HEAD

Diff just staged changes:  
git diff --staged

Diff just unstaged changes:  
git diff

### Diff Commits

Show diff between a commit and its parent:  
git show <commit>

Diff two commits:  
git diff <commit> <commit>

Diff one file since a commit:  
git diff <commit> <file>

Show a summary of a diff:  
git diff <commit> --stat  
git show <commit> --stat

### Ways to refer to a commit

Every time we say <commit>, you can use any of these:

|                   |                  |
|-------------------|------------------|
| * a branch        | main             |
| * a tag           | v0.1             |
| * a commit ID     | 3e887ab          |
| * a remote branch | origin/main      |
| * current commit  | HEAD             |
| * 3 commits ago   | HEAD^^ or HEAD~3 |

### Discard Your Changes

Delete unstaged changes to one file:  
git restore <file>

OR  
git checkout <file>

Delete all staged and unstaged changes to one file:  
git restore --staged --worktree <file>

OR  
git checkout HEAD <file>

Delete all staged and unstaged changes:  
git reset --hard

OR  
git restore --staged --worktree .

Delete untracked files:  
git clean

'Stash' all staged and unstaged changes:  
git stash

### Edit History

"Undo" the most recent commit (keep your working directory the same):  
git reset HEAD^

Squash the last 5 commits into one:  
git rebase -i HEAD~5

Then change "pick" to "fixup" for any commit you want to combine with the previous one

Undo a failed rebase:  
git reflog BRANCHNAME

Then manually find the right commit ID in the reflog, then run:  
git reset --hard <commit>

Change a commit message (or add a file you forgot):  
git commit --amend

### Code Archaeology

Look at a branch's history:  
git log main

git log --graph main  
git log --oneline

Show every commit that modified a file:  
git log <file>

Show every commit that modified a file, including before it was renamed:  
git log --follow <file>

Find every commit that added or removed some text:  
git log -G banana

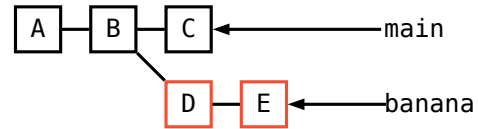
Show who last changed each line of a file:  
git blame <file>

### Combine Diverged Branches

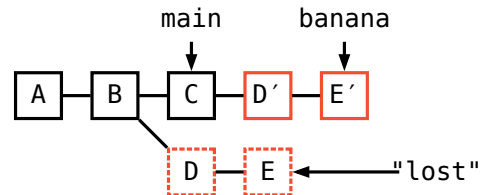
Combine with rebase:

```
git switch banana  
git rebase main
```

Before:



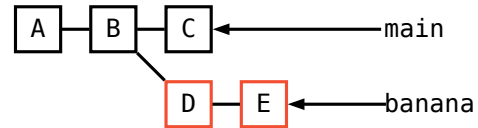
After:



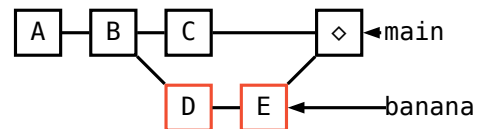
Combine with merge:

```
git switch main  
git merge banana
```

Before:



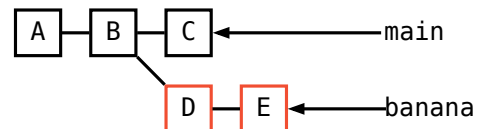
After:



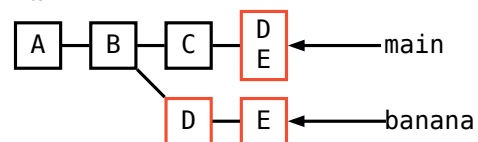
Combine with squash merge:

```
git switch main  
git merge --squash banana  
git commit
```

Before:



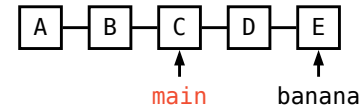
After:



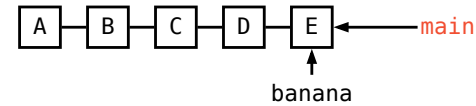
Bring a branch up to date with another branch (aka "fast-forward merge"):

```
git switch main  
git merge banana
```

Before:

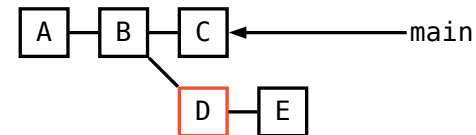


After:

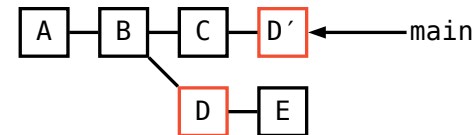


Copy one commit onto the current branch:  
git cherry-pick <commit>

Before:



After:



### Restore an Old File

Get the version of a file from another commit:  
git checkout <commit> <file>

OR

```
git restore <file> --source <commit>
```

### Add a Remote

```
git remote add <name> <url>
```

### Push Your Changes

Push the  
main

branch to the remote  
origin

:

```
git push origin main
```

Push the current branch to its remote "tracking branch":  
git push

Push a branch that you've never pushed before:  
git push -u origin <name>

Force push:

```
git push --force-with-lease
```

Push tags:

```
git push --tags
```

### Pull Changes

Fetch changes (but don't change any of your local branches):

```
git fetch origin main
```

Fetch changes and then rebase your current branch:

```
git pull --rebase
```

Fetch changes and then merge them into your current branch:

```
git pull origin main
```

OR

```
git pull
```

### Configure Git

Set a config option:

```
git config user.name 'Your Name'
```

Set option globally:

```
git config --global ...
```

Add an alias:

```
git config alias.st status
```

See all possible config options:

```
man git-config
```

### Important Files

Local git config:

```
.git/config
```

Global git config:

```
~/.gitconfig
```

List of files to ignore:

```
.gitignore
```